

Presentation Script: Chicago Crime Big Data Analysis

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Part 1: Slide-by-Slide Presentation Script

Slide 1: Title & Introduction

Script: Good morning. My name is Charishma Katta, and my partner is . Our project is titled 'Big Data Analysis of Chicago Crime Records.' Our goal was to demonstrate how to effectively store, process, and analyze a high-volume dataset using a modern, serverless cloud architecture on AWS.

Slide 2: Data Storage & The Data Lake (S3)

Script: We began by creating a Data Lake using Amazon S3. We uploaded our 2.2GB raw CSV file. We chose S3 because it allows for 'Decoupled Storage and Compute.' This is a professional best practice because it lets us store data cheaply without needing a database server running 24/7.

Slide 3: Metadata Discovery & Automation (Glue)

Script: Once the data was stored, we used an AWS Glue Crawler to scan the S3 bucket. This identifies the column headers and data types automatically to populate the AWS Glue Data Catalog, which allows other AWS services like Athena to read the raw files.

Slide 4: Troubleshooting Real-World Data (The Challenge)

Script: This was our biggest challenge. As seen in the screenshot, our initial query returned technical metadata like 'hive' and 'varchar' because the crawler accidentally indexed hidden system files. This required us to pivot to a manual data engineering approach to fix the schema.

Slide 5: Data Engineering & Serialization (SerDe)

Script: To fix the ingestion error, I wrote a custom SQL DDL script using the OpenCSVSerde library. This handles complex CSV text with commas and quotes. I

defined all columns as STRING to ensure 'High Data Availability' and prevent crashes from dirty or empty numeric fields.

Slide 6: Data Validation

Script: We validated the fix with a SELECT query in Amazon Athena. This screenshot proves the fix worked! We can now see actual crime records perfectly aligned in their columns, confirming our Schema-on-Read was correctly mapped.

Slide 7: Outcome 1: Frequency & Density Analysis

Script: We found that 'Theft' and 'Battery' are the most frequent crimes. This analysis identifies where the most pressure is on city resources, helping planners prioritize patrol and administrative efforts.

Slide 8: Outcome 2: Temporal Analysis (Time Series)

Script: We analyzed crime volume from 2001 to the present. This shows the 'Big Data' trend over time, allowing us to see if crime is historically trending up or down, which is essential for long-term policy evaluation.

Slide 9: Outcome 3: Performance Metrics (KPIs)

Script: We calculated the Arrest Rate for each crime type using SQL. This is a key effectiveness metric. We found that Narcotics has a near 100% arrest rate, while others are lower, identifying where law enforcement is most successful.

Slide 10-12: Data Visualizations

Script: These charts (Bar, Line, and Pie) turn our raw SQL results into visual intelligence, showing the volume, trends, and arrest ratios clearly for any stakeholder to understand.

Slide 13: Conclusion: Serverless Analytics

Script: In conclusion, we built a Serverless Pipeline using S3, Glue, and Athena. We processed millions of rows without managing a single server. This architecture is scalable, cost-effective, and robust.

Slide 14: Future Work

Script: For future work, we plan to connect this to Amazon QuickSight for real-time heat maps and use Machine Learning to predict crime 'hotspots' before they occur.

Part 2: Project Understanding (Expected Q&A)

Question: Why did you choose AWS over a local database?

Answer: AWS is serverless and pay-as-you-go. A local database requires managing hardware 24/7. S3 and Athena allow us to scale to petabytes instantly at a fraction of the cost.

Question: Why define columns as STRING?

Answer: This is a 'Schema-on-Read' strategy. The Chicago dataset has 'dirty' data (missing values). Strings ensure 100% ingestion without crashes. We 'CAST' them to numbers during analysis only when needed.

Question: What is a SerDe?

Answer: SerDe stands for Serializer/Deserializer. It is the 'translator' that tells Athena how to turn raw bytes in S3 into the rows and columns seen in SQL.

Question: What would you do if the data grew to 100TB?

Answer: I would implement 'Partitioning.' By organizing S3 data into folders by 'Year' or 'District,' Athena would only scan relevant data, keeping the performance fast and costs low.

Part 3: Project Summary (For Submission)

This project implemented a complete Serverless Big Data Pipeline using Amazon S3, AWS Glue, and Amazon Athena. We successfully ingested 2.2GB of Chicago Crime data, overcame metadata discovery challenges through custom Data Engineering (OpenCSVSerde), and extracted actionable insights including crime density, temporal trends, and arrest rate KPIs. The architecture demonstrates a cost-effective, scalable blueprint for cloud-based analytics.